



QCRI
معهد قطر لبحوث الحوسبة
Qatar Computing Research Institute

جامعة حمد بن خليفة
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- **Arabic podcast** industry is booming.
- Online content is now the best way to reach people, estimates put *global listeners above 500M*
- However, **production is the Bottleneck**
- The traditional podcast production process is slow, expensive, and technically demanding.
- A single episode can take many hours from research to scripting, recording, and editing, which makes consistent publishing hard for small teams.
- It's a significant barrier for new and existing creators.
- AI automation for podcasting addresses these challenges head-on and streamlines the entire workflow from script to voice generation.

System Pipeline



```
graph LR; 1[1. User Input] --> 2[2. Persona Selection]; 2 --> 3[3. Domain Classification]; 3 --> 4[4. Outline Generation]; 4 --> 5[5. Persona Enhancement]; 5 --> 6[6. Cultural Enhancement]; 6 --> 7[7. Final Script Generation];
```

1 User Input
Topic + Information + Speaking Style
Charismatic, Analytical, Friendly, Authoritative, Casual, Enthusiastic, Calm, Persuasive

2 Persona Selection
Host & Guest Matching
Host: Topic + Info
Guest: Topic + Info + Style
Via Similarity Search

3 Domain Classification
Topic Categorization
تقنية صحة تعليم ثقافة أعمال عام

4 Outline Generation
Content Structure Creation
Using Topic + Info + Classification

5 Persona Enhancement
Personality Integration
Speaking patterns, communication styles, role-specific expertise

6 Cultural Enhancement
Cultural Awareness Integration
Arabic cultural context, expressions, regional sensitivities

7 Final Script Generation
Natural Arabic Podcast with Authentic Personas & Cultural Awareness

System Architecture

```
graph TD
    LLM[Large Language Model]
    SS[Speech Synthesis]
    LLM --> SS
    LLM --- LLM_Comp[Large Language Model]
    LLM_Comp --- LLM_Fanar[Fanar LLM for Arabic script generation]
    LLM_Comp --- LLM_CAD[Culturally Aware Persona Database]
    LLM_CAD --- LLM_CAD_Note[Curated collection with OCEAN personality traits and cultural characteristics]
    LLM_Comp --- LLM_WI[Web Interface]
    LLM_WI --- LLM_WI_Note[HTML/CSS/JavaScript frontend with Python backend]
    SS --- SS_Aura[AURA TTS for high-quality Arabic audio]
    SS --- SS_EdgeTTS[EdgeTTS as secondary option]
    LLM_Comp --- SS_Sim[Similarity Search]
    SS_Sim --- SS_Sim_Note[Cosine similarity for persona matching]
    LLM_Comp --- LLM_DC[Domain Classifier]
    LLM_DC --- LLM_DC_Note[6-category Arabic topic classification]
    LLM --- SS --- LLM_DC --- SS_Sim --- LLM_CAD --- LLM_Fanar
```

Large Language Model

Fanar LLM for Arabic script generation

Culturally Aware Persona Database

Curated collection with OCEAN personality traits and cultural characteristics

Web Interface

HTML/CSS/JavaScript frontend with Python backend

Speech Synthesis

AURA TTS for high-quality Arabic audio

EdgeTTS as secondary option

Similarity Search

Cosine similarity for persona matching

Domain Classifier

6-category Arabic topic classification

تقنية • صحة • تعليم • ثقافة • أعمال • عام

```
graph TD; A[AI Podcasting System] --> B[Dual Persona Selection]; A --> C[Cultural Awareness Integration]; A --> D[Multi-Stage Enhancement]; A --> E[End-to-End Automation]; B --> F[• Host: Topic + Information similarity  
• Guest: Topic + Information + Speaking Style]; C --> G[• Arabic cultural context and expressions  
• Regional sensitivities and nuances]; D --> H[• Persona-driven dialogue enhancement  
• Cultural authenticity layer]; E --> I[• From user input to final podcast script  
• No manual intervention required];
```

The diagram illustrates the four main components of the AI Podcasting System, each with specific functionalities:

- Dual Persona Selection**
 - Host: Topic + Information similarity
 - Guest: Topic + Information + Speaking Style
- Cultural Awareness Integration**
 - Arabic cultural context and expressions
 - Regional sensitivities and nuances
- Multi-Stage Enhancement**
 - Persona-driven dialogue enhancement
 - Cultural authenticity layer
- End-to-End Automation**
 - From user input to final podcast script
 - No manual intervention required

Culturally-Aware Generation Personality-Driven Dialogue Domain-Specific Classification Spontaneous Conversation

Methodology

The diagram illustrates the PersonalHub dataset and its application in persona-based dialogue generation. It is divided into three main sections: **Dataset**, **Solution**, and **Spontaneity Techniques**.

- Dataset:** Built from **PersonalHub dataset**, translated to Arabic with cultural adaptation. Each persona includes age, gender, **OCEAN traits**, speaking style, and culturally contextualized job descriptions.
- Solution:** Structured prompting with real code examples initially attempted but proved too prompting-intensive for **Fanar LLM calls**. The solution involves structured prompting with **JSON schema preservation**, cultural templates, and robust fallback mechanisms for LLM failures.
- Spontaneity Techniques:**
 - Natural fillers (كلمات ملأية)
 - Gist dialect integration
 - Emotional tags (<surprised>)
 - Pause markers [pause: 1s]
 - Context-aware dialogue flow

The **Dataset** section includes a box describing the **Database Size**: Extensive collection with deduplication filtering to prevent persona overlap. Generated additional personas using **ChatGPT** for domain-specific coverage.

The **Similarity Search** section describes the host selection via **Topic+Info** vectors: Guest selection via **Topic+Info+Speaking Style** vectors using cosine similarity.

The diagram illustrates the RAG Pipeline Architecture, which consists of four sequential stages:

- 1. JSON Validation:** Strict schema enforcement with structure verification at each pipeline stage.
- 2. Falback Mechanisms:** Rule-based enhancement when LLM parsing fails or returns invalid output.
- 3. Content Validation:** Arabic language correctness and cultural appropriateness checks.
- 4. Error Recovery:** Graceful degradation with conservative fallbacks preserving core functionality.

 Swalf

Automated Arabic Podcast Generation with Cultural Awareness

Transform your ideas into engaging Arabic podcasts using advanced AI. Generate natural conversations, interviews, and storytelling content with cultural authenticity.

Topic / Theme

Renewable energy

Information about the topic

helping reduce greenhouse gas emissions and combat climate change by investing in renewable energy, societies can ensure energy security, create new jobs, and promote a healthier environment for future generations.

Spooking Style

Enthusiastic - متحمس وحماسي

Generate Podcast

[illegible]

We conducted some reviews and forms for feedback from random students where we gave them a sample audio and sample script titled

الطاقة المستقبل

Rating	Number of Respondents	Percentage
1	0	0%
2	6	22.2%
3	9	33.3%
4	8	29.6%
5	4	14.8%

Rating	Number of Respondents	Percentage
1	0	0%
2	6	22.2%
3	7	25.9%
4	12	44.4%
5	2	7.4%

Rating	Count	Percentage
1	1	3.7%
2	2	7.4%
3	7	25.9%
4	14	51.9%
5	3	11.1%

3 responses

المغنى إيشاق فينديروات انشأت من الذكاء الاصطناعي كتساب مع المحتوى

Nice

The podcast is generally good I like that it's culturally aware. However I would prefer if it also have some more real life examples that has happened recently

References

Ju, Zeqian, et al. "MoonCast: High-quality zero-shot podcast generation (2025).
Xiao, Yujia, et al. "PodAgent: A Comprehensive Framework for Podcast Generation." (2025).
Yahagi, Yuchi, et al. "PaperWave: Listening to Research Papers as Conversational Podcasts Scripted by LLM." 2025.
Veluri, Bandhav, et al. "Beyond turn-based interfaces: Synchronous llms as full-duplex dialogue agents." (2024)
Mai, Long, and Julie Carson-Berndsen. "Real-time textless dialogue generation." (2025).

Swalf: Automated Arabic Podcast Generation with Cultural-Awareness

Authors: Abdelrahman Kotb, Ahmed Adam, Jana Abdulmagid

Motivation

- Arabic audio is booming. Podcast audiences keep climbing worldwide and across MENA. Estimates put global listeners above 500M
- Arabic-first AI matters. Arabic content and tooling lag behind. Fanar was launched to strengthen Arabic AI capacity
- Production is the bottleneck. A single episode can take many hours from research to scripting, recording, and editing, which makes consistent publishing hard for small teams.
- Our approach. Combine Fanar LLM for knowledge-grounded, natural Arabic scripts with QCRI's AURA TTS for expressive delivery, creating an agentic system that scales high-quality, culturally resonant Arabic podcasts.

System Architecture & Key Features

System Architecture

Large Language Model
Fanar LLM for Arabic script generation

Speech Synthesis
AURA TTS for high-quality Arabic audio
EdgeTTS as secondary option

Culturally Aware Persona Database
Curated collection with OCEAN personality traits and cultural characteristics

Similarity Search
Cosine similarity for persona matching

Web Interface
HTML/CSS/JavaScript frontend with Python backend

Domain Classifier
6-category Arabic topic classification
تقنية • صحة • تعليم • ثقافة • أعمال • عام

Key Features & Contributions

Dual Persona Selection
• Host: Topic + Information similarity
• Guest: Topic + Information + Speaking Style

Cultural Awareness Integration
• Arabic cultural context and expressions
• Regional sensitivities and nuances

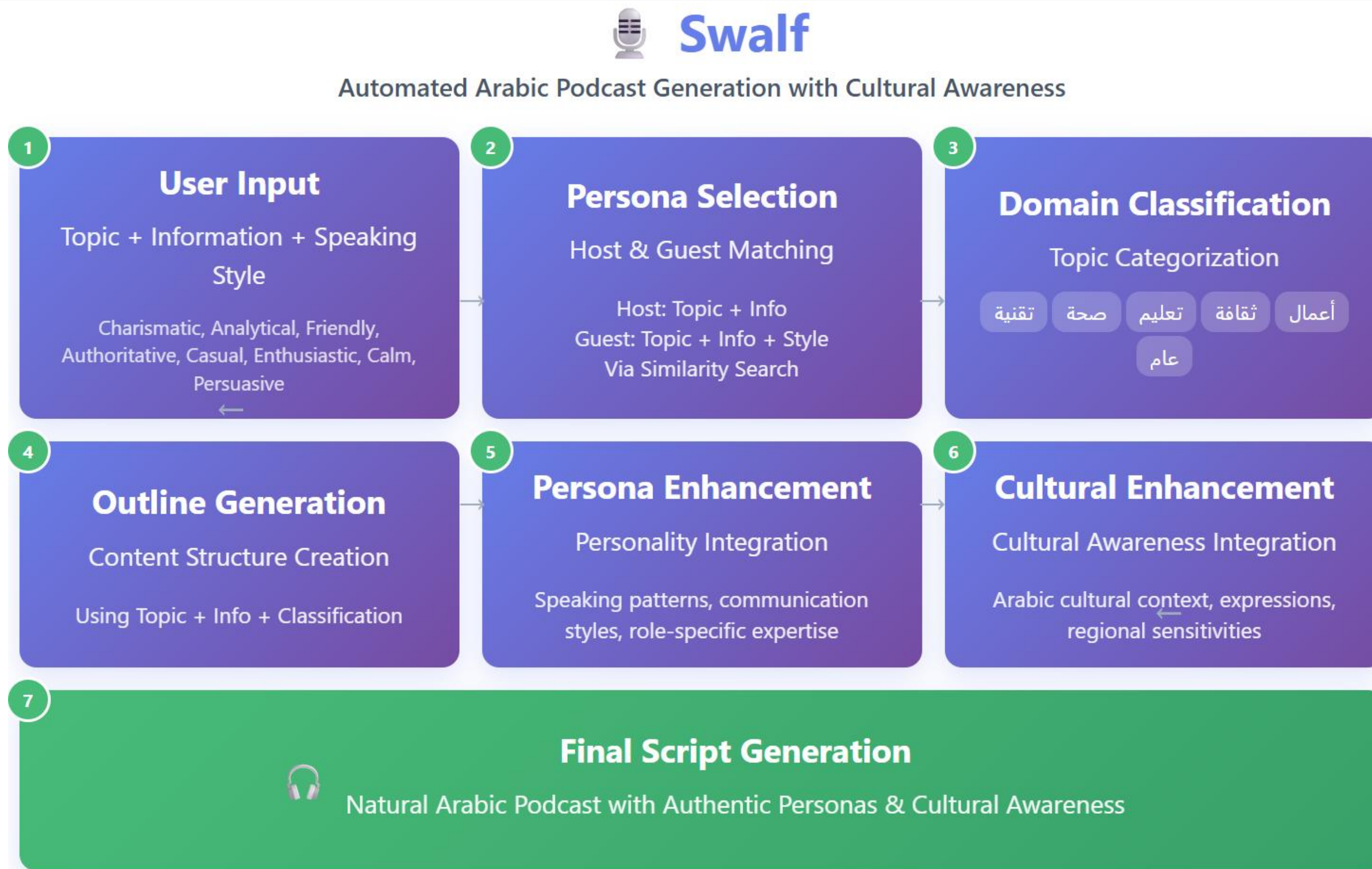
Multi-Stage Enhancement
• Persona-driven dialogue enhancement
• Cultural authenticity layer

End-to-End Automation
• From user input to final podcast script
• No manual intervention required

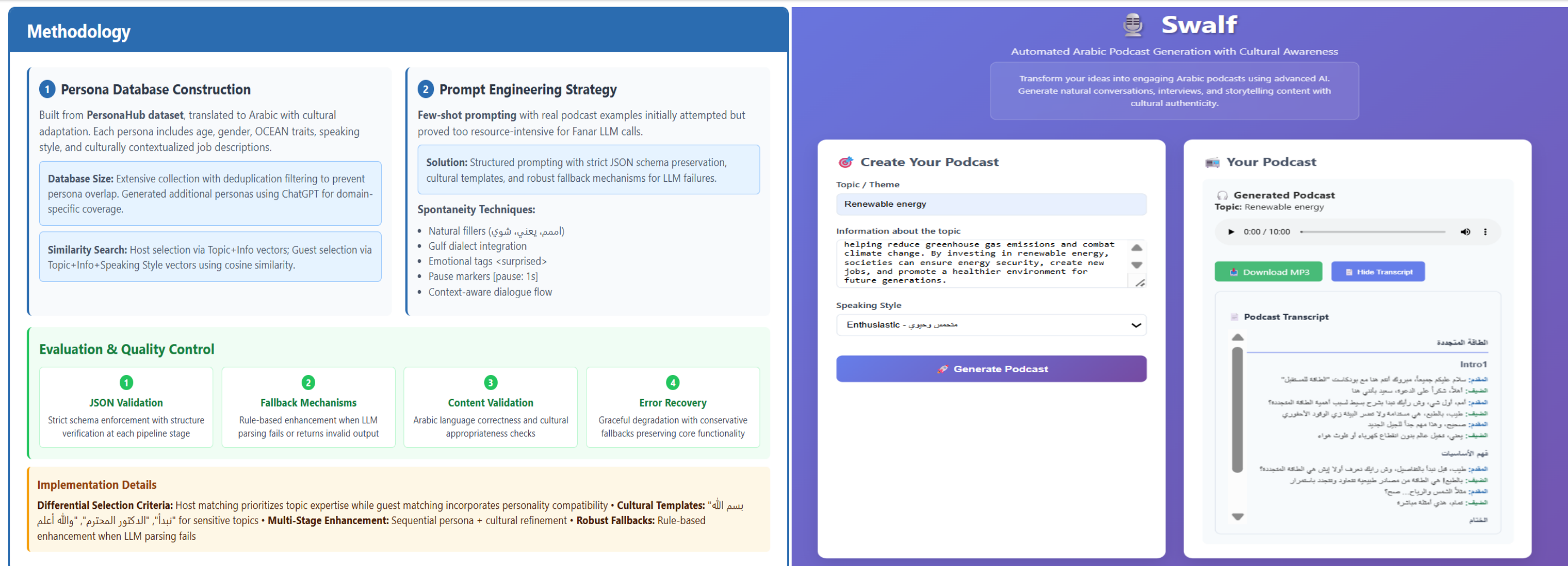
Technical Innovations

Culturally-Aware Generation Personality-Driven Dialogue Domain-Specific Classification Spontaneous Conversation

System Pipeline



Methodology, Web interface & Future Work



Future Work:

Multi-dialect delivery. Add speaker-selectable Arabic dialects (Gulf, Levantine, Egyptian, Maghrebi, MSA)

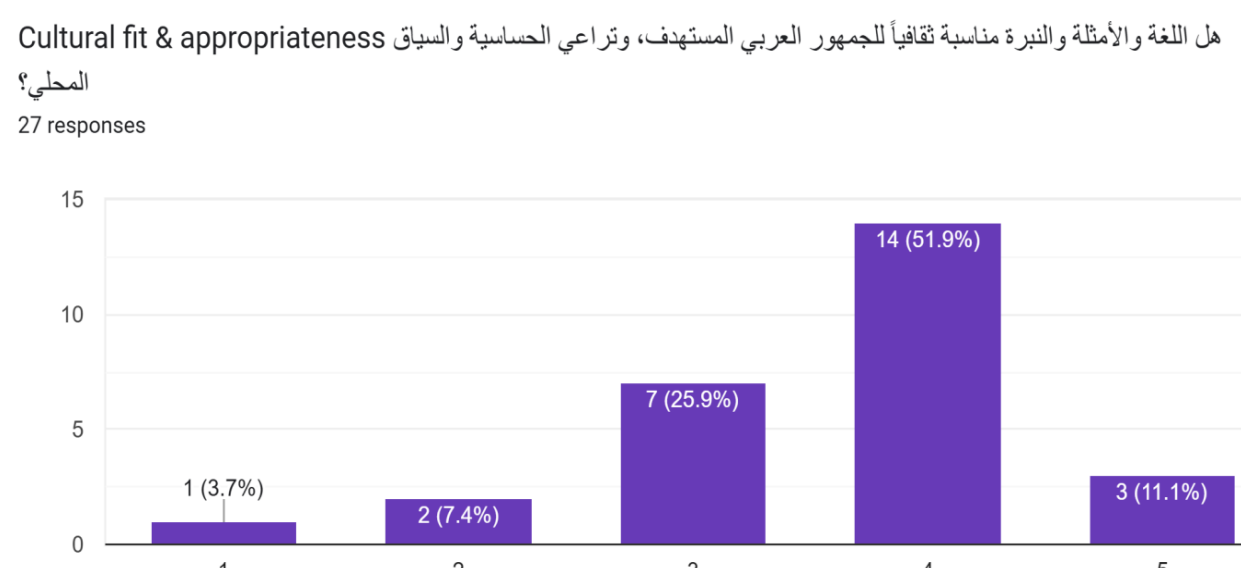
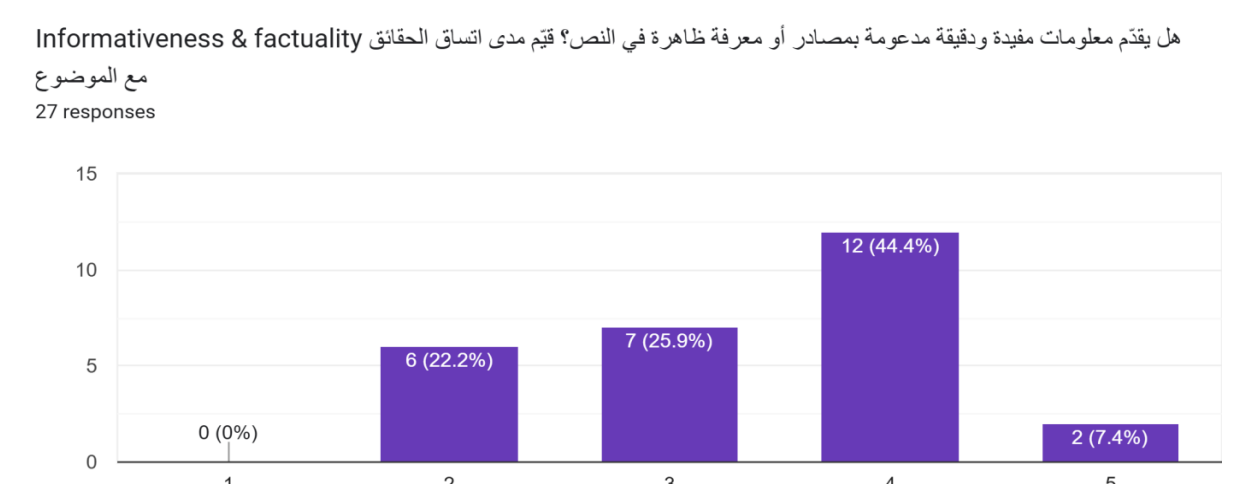
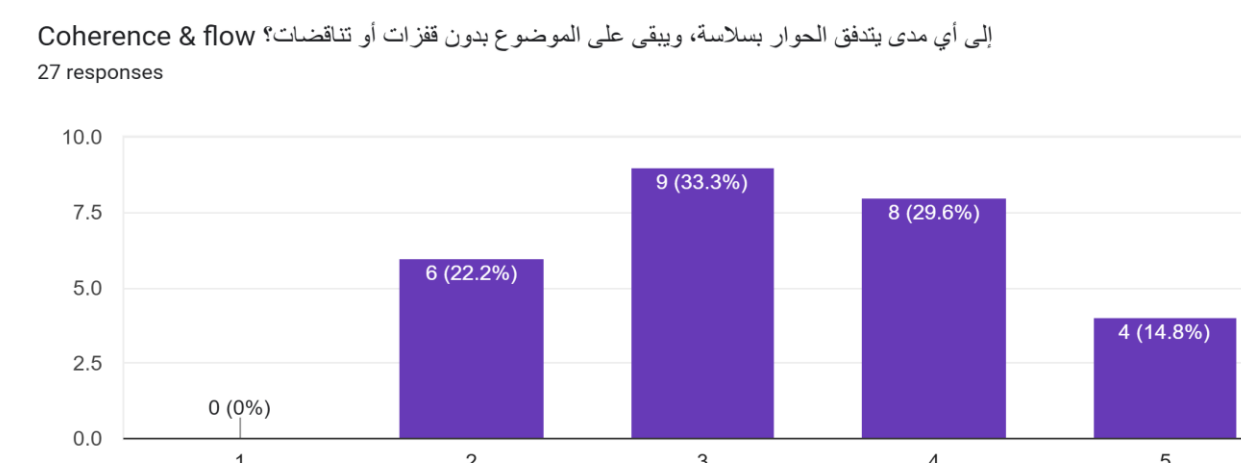
Visual podcasting. Generate video versions of episodes with auto-captions.

Storytelling engine. Introduce narrative templates (hook–conflict–resolution), character roles, and scene

Experiments

We conducted some reviews and forms for feedback from random students where we gave them a sample audio and sample script titled

الطاقة للمستقبل



Extra Comments?
3 responses

أعني إضافة فيديوهات التثقيف من الذكاء الاصطناعي لتتألف مع المحتوى

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